AK

Anirudh Menon Kunnath Pathayapura

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Profile

Enrolled at the University of Toledo, pursuing a bachelor's degree in physics with a concentration in astrophysics and a minor in Data Science , while serving as a Research Assistant at the Wright Center for Photovoltaics Innovation and Commercialization. Known for diligence, approachability, ambition, attentiveness, drive, and strong organizational skills. Passionate about advancing knowledge in Space Photovoltaics and committed to contributing to innovative research projects. Eager to leverage an academic background and hands-on experience to make a meaningful impact in the field.

Work Experience

Wright Center PVIC - Research Assistant
Toledo, OH
02/2024 - Current
Assisted in collecting and analyzing solar cell performance data to identify areas for efficiency improvements. Utilized JV EQE, UV-Vis, Fast Hull, and JMP software to support data analysis efforts.
Contributed to research on ALD-deposited alumina and SWCNT doping for back-surface passivation.

Wilderness exploration club (WEX) - President
Toledo, OH
07/2025 - Current
Organized multi-day outdoor trips (rafting, backpacking, skiing)
Managed logistics, budgeting, and risk planning for 30+ student participants.
Promoted leadership and teamwork through event
RocketHacks - Organizer
Toledo, OH
03/2025 - Current
Contributed to event logistics and experience team by coordinating planning and setup activities.

Projects

My project focuses on improving thin-film solar cell performance using single-walled carbon nanotube (SWCNT) networks. I studied how adding an ultra-thin Al_2O_3 layer using atomic layer deposition (ALD) on SWCNTs can reduce recombination losses at the back interface of CdTe/CdSeTe solar cells. Through device fabrication and characterization (JV and EQE measurements), the results showed positive performance trends, suggesting that Al_2O_3-modified SWCNT back contacts can enhance charge transport and overall device efficiency.

Skills

- Lab Technician
(Wet lab, Sputter systems, Ultrasonic spray systems, Gold deposition)
- Data coding and analysis
(R-Code, JMP, Igor Pro)
- Computer programming
(Python, C++, SQL)
- Researching skills
- Leadership team building

Education

Expected in 05/2027
University Of Toledo
Ohio
Bachelor of Science: Physics With Astrophysics (Minor in Data Science)
12/2023
International Indian School
Dammam, SA
High school Diploma

Languages

English
Hindi
Malayalam
Tamil
Intermediate French

Hobbies

Rock Climbing, Badminton, Guitar